

An Extension of the Sugarscape Model with LLM-based Agents

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Sugarscape is an agent-based model designed to simulate the distributed emergence of social, economic, and demographic phenomena through simple local interactions in a resource-constrained environment. In this study, we extend the classical Sugarscape framework by replacing traditional rule-based agents with Large Language Model (LLM) driven entities. We introduce a multi-dimensional agent architecture featuring heterogeneous metabolism, a kinship proximity metric for social affiliation—based on an α gene parameters—and diverse environmental topographies with varying resource distributions.

The research is designed to compare these LLM-driven agents with baseline rule-based simulations to identify significant departures in resource management, sustainability, and group dynamics, and to assess the presence of emergent behaviors in LLM-based artificial societies. Preliminary findings suggest that LLM-integrated agents develop distinct behavioral patterns that contrast with traditional models, offering a promising foundation for the future modeling of complex artificial societies and their resilience in resource-constrained environment.

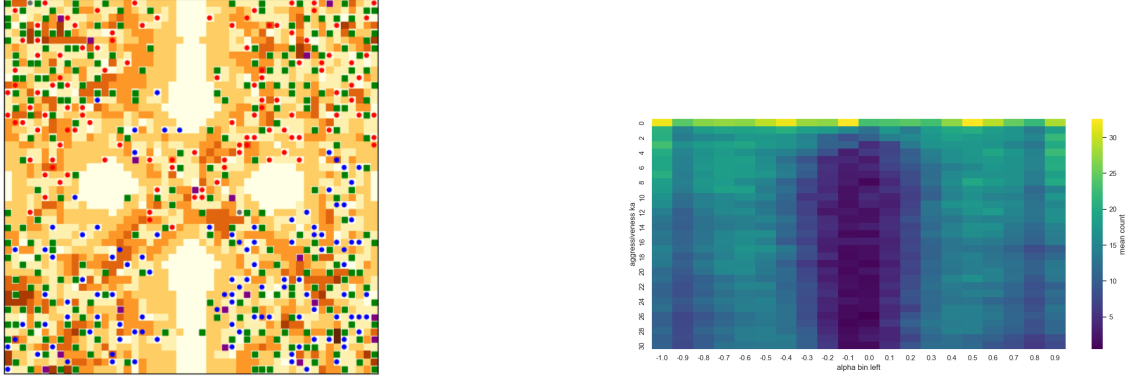


Figure 1: (a) graphical representation of our Sugarscape model (b) heatmap of the aggressiveness—alpha distribution